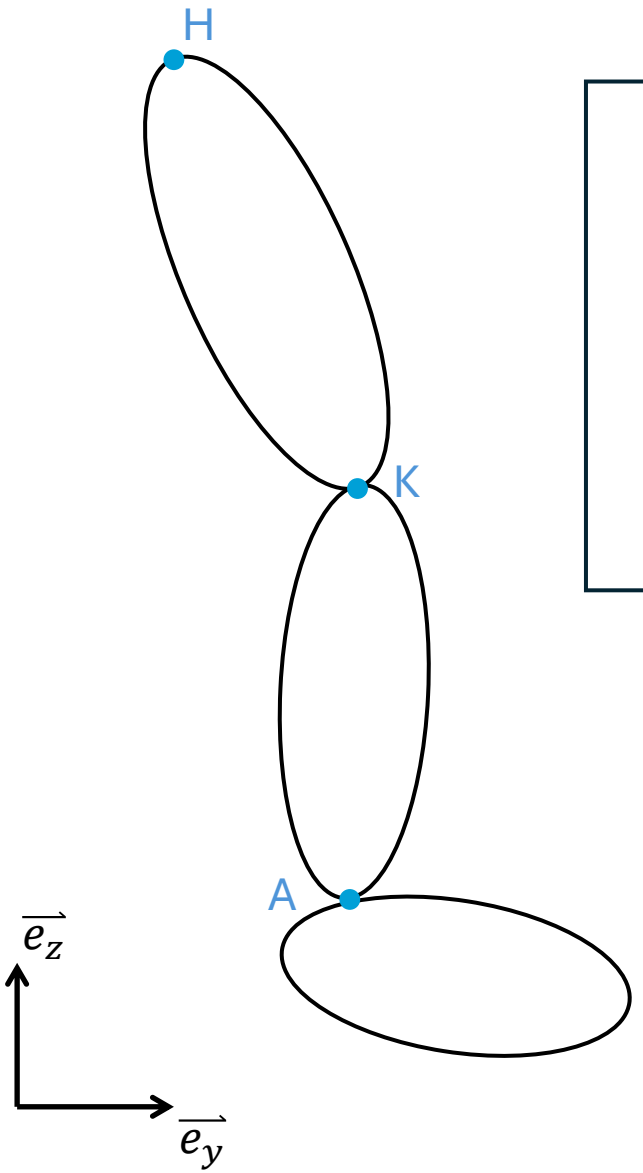


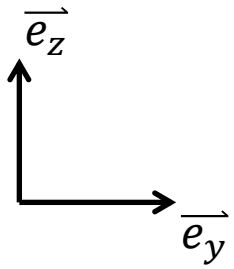
notations:

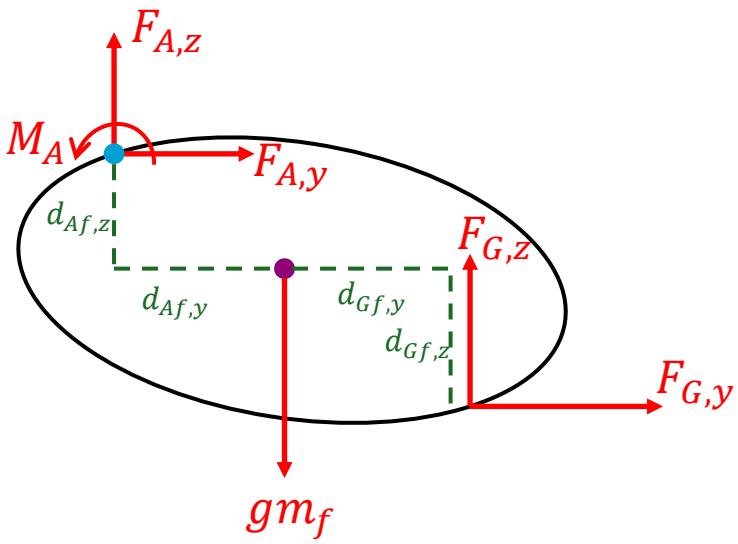
Hip-H	Thigh - t
Knee-K	Shank - s
Ankle-A	Foot - f
Ground-G	

$distance_{point;bodymember,direction}$

Free body foot

- CG:Center of gravity
- Ankle

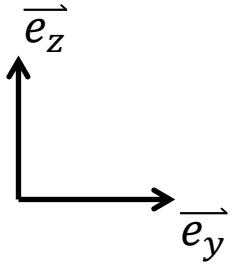


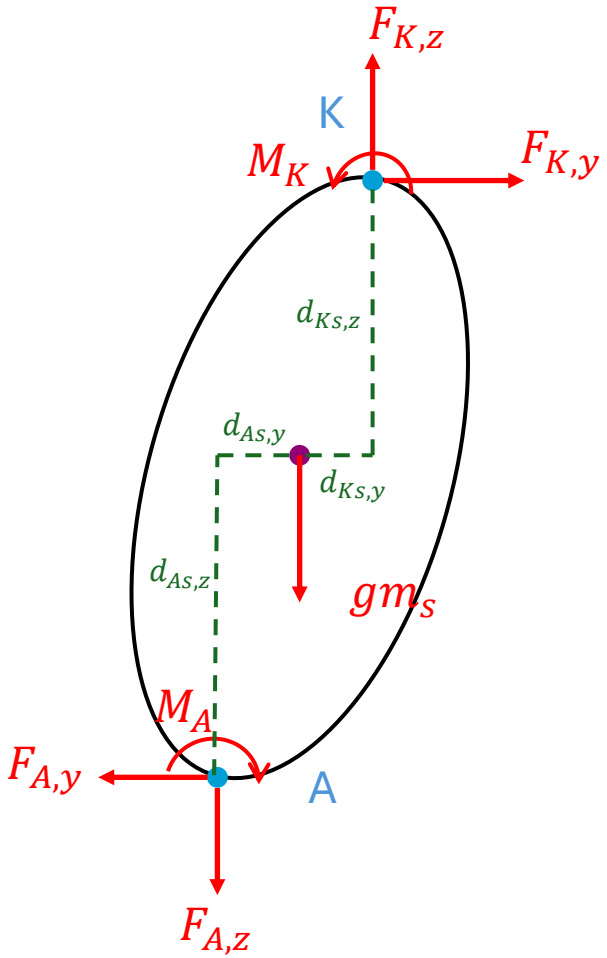


$$\vec{e_y}: \; F_{A,y} + F_{G,y} = m_f \ddot{y}$$
$$\vec{e_z}: \; F_{A,z} + F_{G,z} - m_f g = m_f \ddot{z}$$
$$\overset{\curvearrowleft}{CG}: \; F_{G,z} d_{gf,y} + F_{Gf,y} d_{g,z} - F_{A,z} d_{Af,y} - F_{A,y} d_{Af,z} + M_A = I_{CG,f} \alpha_f$$

Free body shank

- CG:Center of gravity

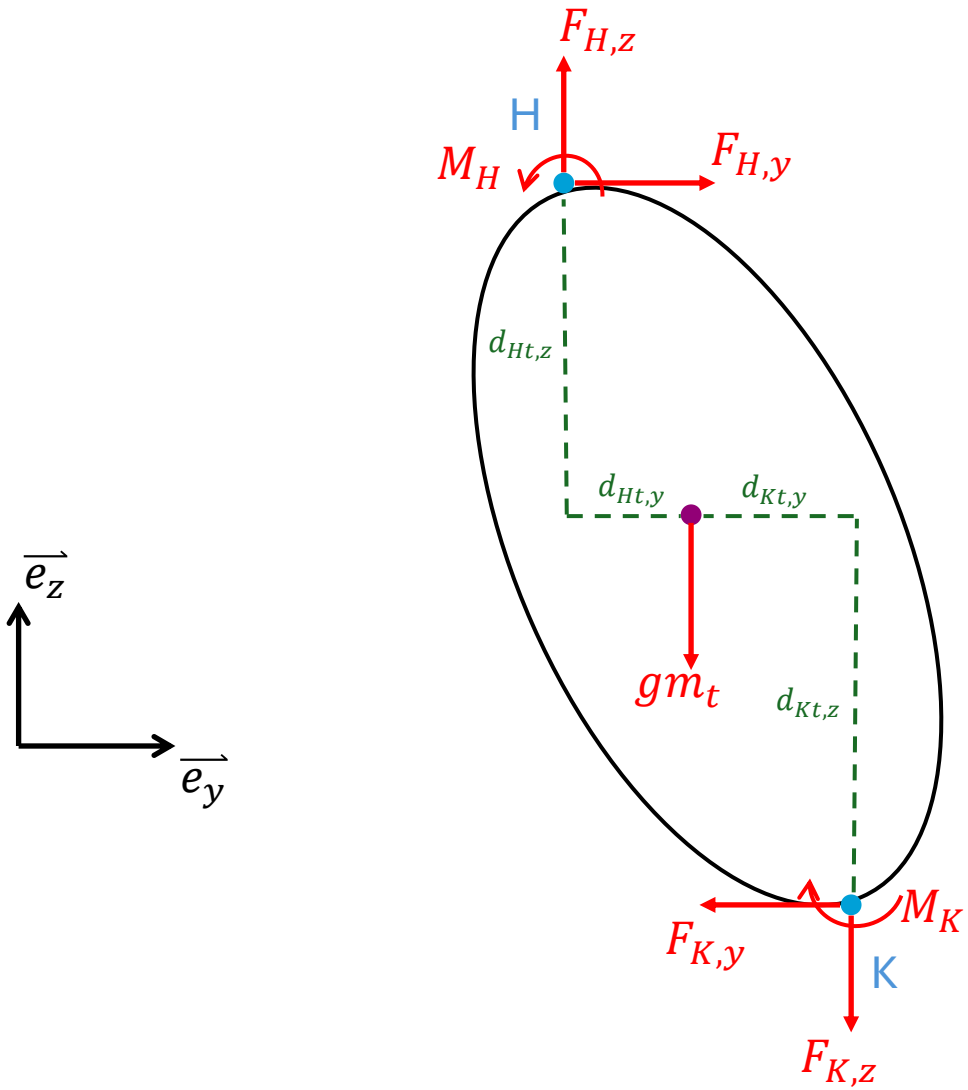




$$\vec{e_y}: \; -F_{A,y} + F_{K,y} = m_s\ddot{y}$$
$$\vec{e_z}: \; -F_{A,z} + F_{K,z} - m_s g = m_s\ddot{z}$$
$$\hat{CG}: \; F_{A,z}d_{As,y} - F_{A,y}d_{As,z} - F_{K,y}d_{ks,z} + F_{K,z}d_{ks,y} + M_K - M_A = I_{CG,s}\alpha_s$$

Free body thigh

- CG:Center of gravity



$$\bar{e_y}: \; -F_{K,y} + F_{H,y} = m_t \ddot{y}$$

$$\bar{e_z}: \; -F_{K,z} + F_{H,z} - m_t g = m_t \ddot{z}$$

$$\hat{CG}: \; -F_{k,z} d_{kt,y} - F_{k,y} d_{kt,z} - F_{H,z} d_{Ht,y} - F_{H,y} d_{Ht,z} + M_H - M_K = I_{CG,t} \alpha_t$$